Data Transformation: Bills of Material (BOM) SAP Business ByDesign to SAP S/4HANA Cloud Public Edition Guide

For SAP Cloud Integration

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1. Introduction

In this guide you shall learn how to consume the integration flow to transform data of object in SAP Business ByDesign into a form suitable and acceptable by SAP S/4HANA Cloud Public Edition

1.1. Definition

This integration flow is an accelerator that transforms the standard data of 'Bills of Material' object in SAP Business ByDesign into the SAP S/4HANA Cloud Public Edition's migration template of object 'Material BOM'.

1.2. Intended Audience

This integration flow is intended to be used by both partners and customers who are in the data migration phase of their implementation project of moving their SAP Business ByDesign system to SAP S/4HANA Cloud Public Edition system.

1.3. Structure

The structure of this guide follows the sequence of steps required to consume the integration flow for the purpose of data transformation using SAP Integration Suite service on SAP BTP.

1.4. System Connectivity

To use this integration flow you don't have to connect your SAP Business ByDesign system or SAP S/4HANA Cloud system to SAP Integration Suite. The data extracted from SAP Business ByDesign is passed to SAP Integration Suite over HTTPS and the data prepared as output is a file that can be taken manually and uploaded in SAP S/4HANA Cloud Public Edition via Migration Cockpit.

1.5. Additional Documentation

You may refer to these topics via links to gain insights into various topics you need to know before using the integration flow.

SAP Integration Suite
https://help.sap.com/docs/cloud-integration/sap-cloud-integration/sap-cloud-integration?version=Cloud

SAP Business ByDesign Data Extraction
https://help.sap.com/docs/SAP_BUSINESS_BYDESIGN/2754875d2d2a403f95e58a41a9c7d6de/2bff33a5722d10148b06d3479e04c151.html?version=2002

SAP S/4HANA Cloud Public Edition Migration Cockpit
https://help.sap.com/docs/SAP_S4HANA_CLOUD/d5699934e7004d048c4801b552f3b013/f32db0c240484241abc53a876253e118.html?version=2408.500

Technical Data Migration SAP Business ByDesign to SAP S/4HANA Cloud
https://learning.sap.com/video-playlists/migrating-data-from-sap-business-bydesign-to-sap-s-4hana-cloud

Setting up Required systems and Users
https://learning.sap.com/videos/preparing-for-data-migration-from-sap-business-bydesign-to-sap-s-4hana-cloud

2. Business Scenario

Customers who are looking to migrate from SAP Business ByDesign ERP system to SAP S/4HANA Cloud Public Edition ERP system need an easy data migration path to save time and effort on data migration activities which is quite intensive.

With this intention SAP has delivered the mapping of data fields from SAP Business ByDesign Bills of Material object to SAP S/4HANA Cloud Public Edition equivalent object as part of this integration flow. The end result of the integration flow will provide you Bills of Material data file that is in the form of SAP S/4HANA Cloud Public Edition's 'Material BOM' migration template. You shall upload this file via Migration Cockpit in SAP S/4HANA Cloud Public Edition. This helps you to save effort and prepare the file easily for upload.

3. Prerequisites

To consume the integration flow you should have the following things done

a) Set up account in SAP BTP and subscribe to SAP Integration Suite service.
b) Set up required user roles and authorization in SAP BTP subaccount
c) Set up the required capabilities in SAP Integration Suite to trigger integration flow.
d) User should have authorisation to the Migration Cockpit app in SAP S/4HANA Cloud Public Edition.
e) Have authorisation to the Data Extraction functionality in SAP Business ByDesign
f) User should have the date format **DD.MM.YYYY** in SAP Business ByDesign system. If the date format is different user shall change it to the said format by navigating to User->My Settings->Date.

Please refer to the links mentioned in the section Additional Documentation to complete the prerequisites.

For setting up the required systems and users you can refer to these tutorials:
https://learning.sap.com/videos/preparing-for-data-migration-from-sap-business-bydesign-to-sap-s-4hana-cloud

4. Consumption of integration flow

Follow the steps below to transform your data and upload to SAP S/4HANA Cloud Public Edition.

4.1. Data Extraction

To consume the integration flow for transformation you need to prepare the data that you want to transform. You should extract the required Bills of Material data from SAP Business ByDesign.

Follow the steps below for preparation. You can refer to the learning session to understand in details:

a. Extract the data from ByD in form of ZIP file:
   i. Login to Business by Design system (where the data is residing) and on the left menu navigate to Application and User Management work centre.
   ii. Choose and open Data Extraction work center view.
   iii. Create a new data extraction run and define the filter parameters as you need.
   iv. Click Activate and Run. This will start the extraction job in the background.
   v. Check the status of the job in the Data Extraction view until it is finished.
   vi. Download the extracted ZIP file.
   vii. Check the extracted file and ensure there are no blanks in any of the columns in the extracted file, ensure to fill the values in place of blanks.
   viii. This file needs to be Zipped again and then passed to the integration flow via Post HTTP request.

b. Create a migration project in S/4 HANA public cloud and make sure you have added the migration object *Material BOM to* the project.

4.2. Data Preparation

No Data Preparation step is needed for the Bills of Material object; the ZIP file extracted from the Data Extraction Work Center view can be used directly.

4.3. Data Transformation

Once the data is prepared and ready in the required form, you are ready to execute the integration flow.

Make the following preparations to load the integration flow for executing data transformation.

a. Login to SAP Cloud Integration and search for the package "**SAP Business ByDesign Master Data to SAP S/4HANA Cloud Public Edition**" in Discover page and open it.
b. Open and copy the package into your workspace.
c. Select the package and click "Copy" to duplicate the entire package, including the "ByD Bills of Material Master Data Transformation" integration flow, along with all other integration flows and documents.
d. Deploy the integration flow. Refer to the learning session for details.
e. After successful deployment of integration flow, make a note of the End Point URL generated. Refer to the learning session for details.

To consume the integration flow for data transformation and create a S/4HANA migration file, follow the steps below. For details refer to the learning session.

a. Set up an application to trigger the integration flow e.g., Postman.
b. Create a POST HTTP request, provide the user credentials, and upload the extracted ZIP file in the body of the request.
c. Send the request to the end point URL of integration flow. This will trigger the integration flow and start the data transformation. As a result, you shall receive a response file back from your triggering application.
d. Save the response ZIP file received. Extract the ZIP package and Open the XML file in Microsoft Excel. Click on *Save*.
e. **Note:** In the response XML, some records contain fields that require your attention. These instances can be identified by locating cells with the value **"<<Enter Manually>>"**.

4.4. Data Upload

Once the data is transformed and you have received the S/4HANA template file filled with the data (response file) you are ready to upload the same to S/4HANA system via Migration Cockpit.

Follow the below steps to upload the data to S/4HANA system:

a. Save the response ZIP file received.
b. Extract the ZIP package and open the XML file in Microsoft Excel. Click on save. Close the file.
c. This is the file that should be taken to SAP S/4HANA Cloud Public Edition migration project and uploaded. For details on uploading a file via migration cockpit in SAP S/4HANA Cloud Public Edition refer to this help document.

5. Customization and Enhancement of integration flow

You can customize the mapping or enhance the integration flow as per your requirement. For details on customization, you can refer to these tutorials:

- Tutorial 1: Customizing standard integration flows for data migration from SAP Business ByDesign to SAP S/4HANA Cloud Public Edition.
- Tutorial 2: Updating data structure in integration flows for data migration from SAP Business ByDesign to SAP S/4HANA Cloud Public Edition.

6. Contact Information

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CA-S4H-BYD-IFLOW